Linear algebra

08 / Matrix products

Dimensions and composition

If A is m by n and B is n by p, AB is m by p. Entry (i,j) is row i of A dotted with column j of B. When A is 2 by 3 and B is 3 by 2, AB is 2 by 2 while BA is 3 by 3. The shared inner dimension permits composition.

Worked matrix product

For C = [2 1; 0 3] and D = [1 4; 2 0], compute each entry by row-column products: 2*1 + 1*2 = 4; 2*4 + 1*0 = 8; 0*1 + 3*2 = 6; 0*4 + 3*0 = 0. Therefore CD = [4 8; 6 0].

Order matters

Matrix multiplication is generally not commutative: AB need not equal BA, and one product may be undefined. Composing transformations in a different order can produce a different transformation. Even when both products exist, their dimensions or entries can differ.

Another product

With A = [1 2; 3 4] and B = [2 0; 1 2], the row-column rule gives AB = [4 4; 10 8]. A useful check is to compute the dimensions before calculating entries, then verify one column as A applied to the corresponding column of B.

Original mock course material for a portfolio demo. Not an institutional handout.

Linear algebra

09 / Systems and subspaces

Span

The span of a collection of vectors is the set of all their linear combinations. Each combination weights the vectors by scalars and adds the results. Two nonparallel vectors in the plane span the plane; two parallel nonzero vectors span only a line.

Linear independence

Vectors are linearly independent when the only coefficients producing the zero vector are all zero. A nontrivial combination that equals zero demonstrates dependence. Independence means no vector in the collection can be expressed as a linear combination of the others.

Elimination example

For x + 2y = 5 and 3x + 4y = 11, subtract three times the first equation from the second to obtain -2y = -4. Thus y = 2, and substitution in the first equation gives x = 1. These values satisfy both original equations.

Null space basis

The null space of A is the set of vectors mapped to zero. For A = [1 2; 2 4], Ax = 0 reduces to x1 + 2x2 = 0. Set x2 = t to obtain x = t[-2; 1]. A basis for the null space is the single vector [-2; 1].

Rank-nullity theorem

For a matrix with n columns, rank(A) + nullity(A) = n. Rank is the dimension of the column space; nullity is the dimension of the null space. For A = [1 2; 2 4], rank is 1 and nullity is 1, summing to the two-dimensional input space.

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